

HEY, WAIT A MINUTE! IF SCALAR IMPLICATURES ARE PRESUPPOSED, WHY DO THEY FAIL THE “*Hey, wait a minute!*” TEST?*

IDO BENBAJI-ELHADAD & OMRI DORON

Massachusetts Institute of Technology & University of Massachusetts Amherst

1 Introduction

It has recently been argued that scalar implicatures (SIs) are presuppositional; e.g. that the inference of the sentence in (1) that not all students passed the test is a presupposition of the sentence (Bassi et al., 2021). The presuppositional approach (henceforth, PA) explains certain puzzling projection patterns of SIs, as well as certain pragmatic effects they trigger.

$$(1) \llbracket \text{Some of students passed the test} \rrbracket = \begin{cases} 1 & \text{if some but not all of the students passed the test} \\ 0 & \text{if none of the students passed the test} \\ \# & \text{if all of the students passed the test} \end{cases}$$

A nagging challenge to PA is that SIs do not intuitively *feel like* presuppositions, and, more concretely, do not pass the *Hey, wait a minute!* test (*HWAM!*), first proposed by Shanon (1976) and discussed more systematically by von Stechow (2004). This test, demonstrated in (2), is based on the observation that in uttering *HWAM!* an addressee can felicitously challenge presupposed, though not asserted, content. However, *HWAM!* also seems to distinguish between SIs and presuppositions; in (3), B can reject the existence presupposition introduced by the definite article in A’s assertion, but B’ cannot similarly reject the SI introduced by *some*.

(2) A: Mary’s aunt is visiting today.

B: Hey, wait a minute! I had no idea Mary had an aunt.

B’: #Hey, wait a minute! I had no idea Mary’s aunt was visiting today. (Pearson 2010)

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- (3) A: Some students passed the test.
 B: Hey, wait a minute! I had no idea you administered a test.
 B': #Hey, wait a minute! I had no idea that not all students passed the test.

We offer a way of thinking about *HWAM!* that renders the observation in (3) unproblematic for PA. To that end, Section 2 introduces PA. Section 3 then introduces some novel assumptions about discourse structure and constraints thereon, and Section 4 presents an account of the *HWAM!* test that rests on these assumptions. Finally, Section 5 shows how the fact that *HWAM!* cannot target SIs can be made to follow from this account, while maintaining the core insight of PA.

2 The presuppositional approach to scalar implicature

Greatly simplifying, a basic result of PA is that sentences like (1), which have a non-weaker alternative (in this case, the *all*-alternative), presuppose that this alternative is false. We abstract away from the details of the mechanism proposed by Bassi et al. (2021) to derive such scalar inferences, and focus here on their presuppositional nature.

Bassi et al. (2021) cite several empirical points in support of PA, which we will not repeat here. Instead, let us present a related observation from Wehbe and Doron (2024), which will be useful for our main point in this paper, as it introduces some notions that will play a role in our argumentation. Some necessary background: We assume a Stalnakerian conception of discourse, in which declarative utterances are aimed at updating the *context set*, the set of all worlds compatible with the common knowledge of discourse participants. In this system, presuppositions impose a constraint on the context set, termed *Stalnaker's Bridge* by von Stechow (2004), demanding that the presupposition of a sentence be entailed by the context for update to succeed (4).

- (4) **Stalnaker's Bridge (SB):** Updating a context set c with an utterance ϕ presupposing p is only permitted if $c \subseteq p$.

Importantly, SB can be satisfied even if the presupposition of a sentence is not common knowledge prior to its utterance, via the process of *presupposition accommodation*. Upon hearing a presuppositional sentence, an addressee can minimally modify the context set so that it entail its presupposition. For example, the sentence *Mary adopted all three of her kids*, which presupposes that Mary has three kids, can be uttered felicitously even if the addressee has no idea how many kids Mary has, presumably by virtue of the addressee's willingness to reevaluate the context set such that it entails that Mary has two kids, before updating it with that presuppositional sentence.

Doron and Wehbe (2022) examine the interaction of SB and another constraint on context update proposed by Stalnaker (1978), termed *Informativity* in (5), which renders utterances whose meaning is entailed by the context set or contradicted by it, infelicitous. They note that the need to satisfy both SB and INF results in presupposition accommodation being blocked whenever the new information carried by an utterance is entirely presupposed: In such cases, accommodation—required to satisfy SB—renders INF unsatisfiable. In other words, accommodation *bleeds* INF. This generalization, pointed out by Orin Percus (in lecture notes) and further discussed by Aravind et al. (2022), is termed *Post-Accommodation Informativity* (6) by Doron and Wehbe (2022).

- (5) **Informativity (INF):** Updating a context set c with an utterance ϕ is only permitted if $c \cap \llbracket \phi \rrbracket \neq c$ & $c \cap \llbracket \phi \rrbracket \neq \emptyset$ (i.e., $\llbracket \phi \rrbracket$ eliminates some but not all worlds from c).

- (6) **Post-Accommodation Informativity (PAI):** A sentence ϕ presupposing p can be uttered felicitously given a context set c only if ϕ is informative relative to $c \cap p$.

PAI can be seen in action in (7a), where the speaker makes it common ground that all of Mary’s kids are adopted, which in turn makes (7a) equivalent to its presupposition that Mary has three kids. Importantly, when the information that Mary has two kids is asserted rather than presupposed, the result is felicitous (7b). Wehbe and Doron (2024) show that SIs are subject to the same PAI-effects. For instance, in (8), once it is common ground that at least some of the students passed the test, a sentence which conveys that not all passed via a SI is infelicitous. This is unexpected if the *not-all* SI is a part of the asserted meaning of the sentence, since the information that not all students passed is new, and should therefore save the sentence from an INF violation. Indeed, the variant in (8b), which asserts that not all students passed, is felicitous. PA allows us to explain the infelicity in (8a) as a PAI-effect: Accommodation of the *presupposition* that not all students passed renders the sentence uninformative.¹

- (7) **Context:** The addressee does not know how many kids Mary has. The speaker says: I knew that Mary adopted all of her kids, but today I learned something interesting. . .
- a. #. . . she adopted all three of her kids.
 - b. . . . she has three adopted kids. (adapted from Doron and Wehbe 2022)
- (8) **Context:** A professor is talking to a colleague about the results of the test in a course she teaches. Mary and Jane are among the students in that course. She says: I knew that Mary and Jane passed the test, but today I learned something interesting. . .
- a. #. . . some of the students passed the test.
 - b. . . . only some of the students passed the test. (adapted from Wehbe and Doron 2024)

3 Constraints on QuD selection

In Benbaji-Elhadad and Doron 2024 we suggest that an independent observation regarding accommodation may similarly be explained as a bleeding relation between accommodation and independent pragmatic constraints. This is the observation, dating back at least to Lewis (1979), that we seem reluctant to accommodate “surprising” or “controversial” information. While these notions are often not formally defined, the examples in (9) are illustrative.

- (9) a. Daughter to her father, who does not know she is engaged: O dad, guess what?
#My fiancé is subleasing my apartment this summer. (adapted from von Stechow 2008)
- b. Sorry I’m late, #I was parking my camel. (adapted from Kissine and Pantazi 2020)

Our account of this effect requires some further exposition: As a first step, we adopt a version of the Stalnakerian model of discourse enriched with *Questions under Discussion* (QuDs). In the enriched model, every declarative utterance is uttered not only relative to a context set, but also relative to a partition of this set to equivalence classes, *cells*, intuitively corresponding to the

¹We note that this pattern is also predicted by Gricean theories of SIs. See Wehbe and Doron (2024) for a discussion of this and further evidence for PA.

question which discourse participants aim at answering (Roberts, 1996/2012). The QuD affects the felicity conditions of utterances by imposing a *Relevance* constraint, given in (10).²

- (10) **Relevance (REL):** Given a QuD Q , updating context c with an utterance ϕ is only permitted if $\exists Q' \subseteq Q$ s.t. $\llbracket \phi \rrbracket \cap c = \bigcup Q'$ (i.e. $\llbracket \phi \rrbracket$ does not cut through cells in Q).

Recall that INF, introduced in (5) above, requires that every utterance eliminate some but not all worlds from the context set. REL demands that utterances maintain the integrity of each cell in the QuD. Taken together, the derived constraint INF+REL demands that every utterance be equivalent to a union of a *proper* subset of cells. This is because, if some but not all worlds must be eliminated, and no distinction can be made among worlds in the same cell, then effectively, some but not all cells must be eliminated.³ In (11) we define the notion of *good answerhood*; a property of that a sentence has given a context and a QuD, if it satisfies this constraint.

- (11) ϕ is a *good answer* to QuD Q given a context c , only if ϕ satisfies INF+REL relative to Q, c .

A second step towards an account of (9) involves observing that in certain cases accommodation bleeds *good answerhood*. These cases, observed by Irene Heim (class notes), consist of sentences whose presupposition, when accommodated, leaves only one cell in the QuD. After accommodation, such sentences cannot satisfy (11): They satisfy REL only if they either eliminate the entire remaining cell, or leave it entirely intact; either way, however, they end up violating INF. This leads to infelicity. For instance, A's question in (12), which presumably is taken to be the QuD, partitions the context into two cells: (i) those in which Mary has at least one sibling, and (ii) those in which she has none. B's sentence presupposes that Mary has a sister, and so it requires an accommodation step prior to context update that eliminates the second cell in order to satisfy SB. Accommodation leaves us with a one-cell QuD, and a relevant answer to it will necessarily be uninformative; hence the infelicity of B's response.

- (12) A: Does Mary have any siblings?

B: #She has to pick up her sister from the airport. (adapted from Heim, class notes)

Example (12) contains an overtly-uttered question, assumed to be the QuD. In general, however, determining what is the QuD at a given point in discourse requires complicated coordination among interlocutors. As a final step towards an account of (9), we propose that this coordination process is constrained; i.e., that some relation R imposes an ordering (possibly partial) on the set of possible QuDs, that orders them according to how *urgent*—in an intuitive sense—their resolution would be to discourse participants. We then posit the *Urgency Principle* (UP) in (13) that at any given point in discourse, constrains the possible question that the speaker may choose as her QuD. In the final section of the paper, we will spell out a minimal condition that this notion of *urgency* has to meet. However, our ultimate goal is to illustrate how accepting an urgency-based ranking of this sort pays off, both in the context of the theory of presupposition accommodation and in the context of the presuppositional approach to scalar implicature, even without committing to a detailed account of urgency.

²This is the formulation of Relevance proposed by Lewis (1988), but it is not the only one on the market. See Benbaji-Elhadad and Doron (2023, 2025) for discussion. Note that the central point of this paper, explaining why SIs do not seem to pattern like presuppositions with respect to the *HWAM!* test, will turn out to rely on the Lewisian formulation being the one that is posited by the pragmatics. To the extent that our proposal is convincing, it can thus be adduced to support the Lewisian approach.

³In fact, it demands something stronger: that at least one cell be eliminated, and that remaining cells be left intact.

- (13) **Urgency Principle (UP):** Given an ordering R , a context c , and a set $\bar{Q}$ of possible partitions of c , a speaker may not address a question $Q \in \bar{Q}$ if there exists a higher-ranked question given R to which she can provide an answer.

Here is how this brings us closer to an explanation of the pathological cases of presupposition accommodation in (9): Imagine that R ranks some question Q higher than some other question Q' . In that case, a speaker is required to address Q before addressing Q' , if she can provide an answer to both. If the speaker utters a sentence ϕ , whose presupposition p resolves Q and whose asserted content resolves Q' —thereby indicating that, indeed, she is knowledgeable with respect to, and can thus provide an answer to, both of them—then Q , rather than Q' , must be the QuD that her utterance attempts to address. In that case, accommodation of p will make ϕ unable to satisfy INF+REL relative to Q , ruling out ϕ as a good answer to Q . Then, once it is assumed that the questions the speaker is forced to address when uttering the infelicitous sentences in (9)—given our stipulated ranking—are exactly those that are resolved by the presuppositions of these sentences, the infelicity in (9) and (12) have the same, relatively well-understood source; namely, constraints on good answerhood, which are violated in both cases.

More concretely, consider (9a), in which a father is conversing with his daughter who—unbeknownst to him—is engaged to be married. Since the daughter is in a position to answer both *did you get engaged?* and *what will you do with your apartment in the summer?*, she is required to answer the higher-ranked among these two questions. Assuming that the question of the daughter’s engagement is more urgent than a question about her subleasing plans, which seems to us rather intuitive, UP demands that the daughter resolve that question first. However, once the presupposition of (9a), that the daughter is in fact engaged, is accommodated, the sentence does not constitute a good answer relative to the engagement question, resulting in infelicity.

This account of the oddness of cases like (9) gives rise to a prediction that a similar oddness should be observed even when the information conveyed by these examples is “repackaged” in a way that no longer requires that interlocutors accommodate any controversial information. For instance, it would have been equally odd for the daughter in (9a) to utter (14) to her father. As we note in Benbaji-Elhadad and Doron 2024, this remains unexplained if the oddness of (9a) is due to some constraint that bans *accommodation* of controversial information. On the other hand, it follows from UP which demands the daughter address questions in a particular order.

- (14) O dad! I’m subleasing my apartment this summer. #Also, I got engaged to the subleser.

Another benefit of tracing the oddness of (9) to violations of conditions on good answerhood, rather than to a constraint on presupposition accommodation, has to do with the architecture of our model of discourse. In the Stalnakerian model, the availability of accommodation obscures the empirical difference between presupposition and assertion, given that both accommodation and context update by assertion result in a modified context set. As has been pointed out (by e.g., Gauker 1998, Szabó 2006, Simons et al. 2010 a.o.) this renders quite mysterious those pathological cases of accommodation in (9): Why should we observe a difference between the information that we are allowed to assert and the information that we are allowed to presuppose, if there are mechanisms in place to allow both kinds of information to be incorporated into the context set? The answer suggested by our proposal here is, simply, that there is no difference; in principle, anything that can be asserted can be presupposed and accommodated. There are, however, certain demands from good answers that are unsatisfiable after certain information has been successfully accommodated, leading to infelicity. On this view, the process of reconsidering a context set

to satisfy the constraints discussed above—particularly, SB (4) and good answerhood (11)—is triggered automatically. It is only when no such reconsideration leads to a context that can satisfy *both* constraints that we sense infelicity.

What is crucial for the success of such an account of (9) is that the principle dictating the order in which information is to be introduced into discourse—namely, UP—be formulated as a constraint on the order in which open *questions* are to be addressed. It is this feature of the account that traces the cases in (9), in which a background question is stipulated, and those in (12), in which a question is overtly posited, to the same source. This feature of the model will help us explain the inability to resist scalar inferences (SIs) with *HWAM!* even if SIs are presupposed. But first, we must say a few things about the *HWAM!* test itself.

4 A new approach to *HWAM!*

Given our view of presupposition accommodation as constrained only indirectly by constraints on context update and QuD selection, one might wonder how felicitous *HWAM!* responses to presuppositional utterances like those in (2) can be accounted for. When a discourse participant protests someone's utterance via *HWAM!*, they presumably do so because they are under the impression that the utterance violated some pragmatic constraint. But if there are no principles that directly constrain accommodation, what is the principle whose violation *HWAM!* responses protest? We propose to think of *HWAM!* as a way for a discourse participant to protest the speaker's choice of QuD. On this view, *HWAM!* is simply a way for an interlocutor to coordinate the ranking of QuDs with the speaker, by indicating that the ranking they had in mind is distinct from what is betrayed by the speaker's utterance.

More particularly, an addressee B may protest with *HWAM!* in response to a sentence uttered by a speaker A if that sentence indicates that A entertains a ranking of potential QuDs which is unambiguously different from what B has been assuming. One might think that this is the case whenever A's utterance is not a good answer to the most urgent question in the ranking assumed by B, indicating that the utterance is aimed to address a different question. But this is not enough. Since UP prohibits a speaker from answering a particular question when there is a more urgent one to which she can provide an answer, it may be the case that A's ranking is the same as B's, and the reason she does not provide an answer to the most urgent question is simply because she does not have any information that would constitute a good answer. To license a *HWAM!* response, it should therefore also be the case that A's utterance indicates that she is able to answer a more urgent question (by B's ranking) than the one she actually answers. In other words, A's utterance should give rise to an inference which is itself a good answer to a more urgent question, e.g. by presupposing it. This is formulated in (15).

- (15) **The *HWAM!* generalization:** Responding to a declarative sentence *S* uttered relative to a QuD *Q* by asserting "*HWAM! I didn't know that S'*" is acceptable only if
- i. by uttering *S* a speaker indicates that she believes *S'*;
 - ii. there exists a potential QuD *Q'*, more urgent than *Q* given a ranking of possible QuDs;
 - iii. *S'* is a good answer to *Q'*; and
 - iv. *S* is not a good answer to *Q'*.

To see how this works, consider (2) again, where B’s *HWAM!* response signals that she thinks the question *does Mary have an aunt?* (or any other question to which the presupposition of A’s utterance constitutes a good answer) is more urgent than the question that A actually addresses (presumably *who is visiting today?*). By presupposing an answer to the more urgent question, A betrays that she is knowledgeable with respect to it, and is therefore bound by UP to address it first, given B’s ranking of QuDs. To indicate this apparent violation of UP, B responds with *HWAM!*

On this view, the controversiality cases in (9) are edge cases of the general *HWAM!* pattern. The *HWAM!* test relies on our ability to imagine a context in which one has in mind a potential QuD to which the presupposition of an utterance is a good answer, and which is more urgent than the QuD addressed by the assertion. In cases of infelicity due to a controversial presupposition, not only can we imagine such a context, it is hard to imagine any context with the opposite property; namely, a context in which one has in mind a QuD that the assertion might address and which is more urgent than any question relative to which the presupposition is a good answer. These are thus cases in which any reasonable addressee should respond with *HWAM!*

If all this is on the right track, we expect to cases that are a mirror image of (9); namely, examples in which it is hard to imagine a potential QuD to which the presupposition of an utterance is a good answer, and which is more urgent than any reasonable question that the utterance itself might address. Such an example is given in (16a) below, which presupposes the asserted content of (9a) and asserts its presupposition. Given our intuition that the engagement question is more urgent than the subleasing question across different contexts, it comes as no surprise that a *HWAM!* response by the dad in (16a) is degraded. Indeed, even if the dad had no clue that the daughter was planning to sublease her apartment, the importance of the question addressed by the utterance seems to prevent him from protesting. Compare that to a *HWAM!* response to the original example (16b), which is of course completely natural.

- (16) a. Daughter: O dad, guess what? I got engaged to the person subleasing my apartment.
Dad: #Hey, wait a minute! I didn’t know you were subleasing your apartment!
- b. Daughter: O dad, guess what? My fiancé is subleasing my apartment.
Dad: Hey, wait a minute! I didn’t know you got engaged!

5 Why *HWAM!* cannot target SIs

Finally, we can turn to our primary goal; namely, reconciling the presuppositional approach (PA) to SIs with the fact that they do not pass the *HWAM!* test (17), and in general do not give rise to the same feeling of extra effort which is often felt when accommodating presuppositions.

- (17) a. A: Some students passed the test.
b. B: #Hey, wait a minute! I had no idea that not all students passed the test.

Within our approach to *HWAM!*, there is a fairly natural assumption we could make about our urgency ranking on questions that, together with a commonly-accepted view of SI-generation, would explain these facts. This view of SIs-generation, originating with Horn 1972 and defended extensively by Magri (2009, 2011), suggests that in the process of generating SIs, non-relevant alternatives are filtered out. Thus, whenever SIs are generated, they are necessarily relevant to the QuD addressed by the sentence whose utterance generates them. This implies that in (17a), the

not-all implicature should arise only if the alternative *all of the student passed* is relevant relative to the presumed QuD. Indeed, it has long been noticed that when the context is set up to highlight a QuD like *did some of the students pass?* which renders the *not-all* alternative irrelevant, a SI does not arise. In this discussion we can ignore such cases, since, when the *not-all* inference does not arise, attempting to deny such an inference with *HWAM!* does not make sense. We thus assume that the SI is relevant in all the examples considered here.⁴

Assuming PA, this means that the presupposition of, e.g., (17a) is always relevant (given the context set before presupposition accommodation). This, in turn, entails that the QuD addressed by (17a) has a particular shape, described in (18).

- (18) The QuD Q addressed by the speaker in (17a) is such that $\exists Q_1, Q_2, Q_3 \subset Q$ such that:
- i. $\bigcup Q_1 = \{w : \text{none of the students passed in } w\}$;
 - ii. $\bigcup Q_2 = \{w : \text{some but not all of the students passed in } w\}$; and
 - iii. $\bigcup Q_3 = \{w : \text{all of the students passed in } w\}$.

In other words, each of the propositions in (18i-iii) must be a cell or a union thereof in the QuD. More generally, given a sentence which gives rise to a SI, both the assertion of the sentence and each of its negated alternatives must constitute a cell or union thereof of the QuD; the alternatives—as otherwise, they would not be drawn in the first place, and the assertion—as otherwise the sentence would not constitute a good answer to its respective QuD.

Now let us consider what would need to be the case for *HWAM!* to felicitously target the SI drawn in (17). According to (15), there would need to be an alternative question Q' which is more urgent than any question that meets the requirements in (18), and to which the *not-all* SI (but not the asserted content of (17a)) is a good answer. This means that Q' must contain a cell (or union thereof) contextually equivalent to the *all-cell* in (18iii). Interestingly, any question Q' which satisfies this demand stands in a particular logical relation to any question Q that satisfies (18): It is a *subquestion* of Q , where subquestionhood is defined as in (19), following Roberts (1996/2012: ex. 10iii).⁵ This is because any full answer to Q' will determine whether all of the student passed, or not, which inevitably constitutes a partial answer to Q .

- (19) Q' is a subquestion of Q iff any full answer to Q' entails a partial answer to Q .

It is therefore natural to stipulate the following property of the ranking of potential QuDs underlying the urgency principle in (13): If a question Q' is a subquestion of Q , then Q' cannot be more urgent than Q . This immediately leads to the consequence that *HWAM!* can never target SIs,

⁴Bassi et al. (2021) consider the possibility that it is exactly because we may take an alternative to be irrelevant and thus not generated, that *HWAM!* cannot target SIs; e.g., that upon hearing (17a), an interlocutor reluctant to accommodate a potential *not-all* inference may simply assume that the *all*-alternative that would generate it is irrelevant and thus that the inference was not generated. This, they suggest, would be “less disruptive to the flow of conversation” than a *HWAM!* response. On this approach, the only way to object to a *not-all* inference would be to assume that it was not drawn. We think this result is unsatisfactory: *HWAM!* seems unable to target SIs even when a *HWAM!* response is significantly *less* disruptive than assuming a SI was not generated. For instance: A: *Why is John sad?* B: *In Mary’s class all of the students passed, but in HIS class, some did.* A: *#HWAM! I didn’t know that not all of his students passed.* Here, the inference that not all of John’s students passed explains why he is sad, and must be drawn for B’s response to A’s initial question to make any sense, and yet A’s *HWAM!* response is still infelicitous.

⁵For a useful discussion of Roberts, see (*a.m.o.*) Büring 2003, Onea 2016, Velleman and Beaver 2016, Hénot-Mortier 2025.

since any question to which the SI is a good answer is a subquestion of the QuD relative to which the SI-introducing sentence has been uttered.⁶

Interestingly, *subquestionhood* is the relation used by Roberts (1996/2012) to define what she calls the “strategy of inquiry” interlocutors may apply when attempting to reach full resolution of a given QuD: Once a particular QuD is posed, interlocutors may raise coarser-grained subquestions, as intermediate steps towards fully resolving it. Our stipulation above can be recast within Roberts’s model as the claim that a question that can be part of a strategy of resolving some QuD, cannot possibly be more urgent than the QuD whose resolution it aims to advance.

On the view we propose here regarding (17), B is required to accommodate the SI generated for A’s utterance. However, given that accommodation is a transparent process that is triggered automatically, no extra effort is felt on B’s part. Furthermore, accommodation of this presupposition does not lead to a violation of good answerhood, and therefore B cannot use *HWAM!* to resist it. Notice that this reasoning crucially relies on the idea, discussed above, that SIs must be relevant. This sets SIs apart from other presuppositions—e.g., the presupposition that the daughter is engaged, introduced by the definite description “*my fiancé*” in (9a)—which do not have this property. The argument here thus does not rule out *HWAM!* in the general case, which is of course a desired result.

6 Conclusion

We offer an explanation for the infelicity of *HWAM!* responses targeting SIs, couched in an approach to QuD selection in which a ranking on potential QuDs regulates which question interlocutors may address at any given point in conversation. This ranking, in turn, interacts with informativity and relevance constraints in a way that, we argue, accounts for the apparent difficulty of accommodating certain presuppositions, and for the distribution of felicitous *HWAM!* responses. On our analysis, the property of SIs that bars them from constituting the target of *HWAM!* is a consequence of the independently-motivated assumption that SIs are only triggered if they are relevant, which sets them apart from other, non-scalar presuppositions. Connecting these two properties is done via an additional assumption, constraining the possible ranking of QuDs, which builds on observations in Roberts 1996/2012. This assumption is particularly natural given an approach to QuD selection of the kind we advocate for here, lending support to the overall package we propose.

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⁶Notice that in this case, the subquestionhood relation is symmetric: *Q* must also be a subquestion of *Q'*. This means that neither can be more urgent than the other.

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